Kevin Tieu

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Education

Virginia Polytechnic Institute and State University (Virginia Tech) – Blacksburg, VA **expected May 2028**
Bachelor of Science in Aerospace Engineering Major GPA: 3.88/4.00
- *Dean's List x3 (with Distinction x1)* Cumulative GPA: 3.62/4.00

Skills

CAD: Creo Parametric, Siemens NX, Fusion 360, SolidWorks
Simulation & Analysis: STAR-CCM+, OpenVSP, ANSYS Fluent, JMARS, RASAero II, OpenRocket, Simulink
Programming: Python, MATLAB, Linux (Ubuntu), Raspberry Pi, Arduino, LaTeX

Relevant Experience

CoAspire, LLC **May 2026 - August 2027**
Aerospace Engineering Intern May 2026 - August 2026
- Conducted RANS CFD analyses in STAR-CCM+ for missile and diverterless supersonic inlet configurations, evaluating 6-DOF aerodynamic coefficients, shock behavior, flow separation, mass flow, and pressure recovery.
- Developed Python programs and Java macros to automate CFD setup, data extraction, and boundary condition control, reducing manual simulation time by 20%.

Aerospace Engineering Intern May 2027 - August 2027
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SPA Hypersonic Alternative Navigation Research, Virginia Tech Hume Center **October 2026 - Present**
Undergraduate Research Fellow
- Investigating alternate navigation methods for hypersonic vehicles through six-degree-of-freedom (6-DOF) flight dynamics modeling, sensor integration, and trajectory analysis.
- Analyzing optical sensor performance and uncertainty propagation to evaluate navigation accuracy, measurement sensitivity, and robustness under hypersonic flight conditions.

Orbital Launch Vehicle Team at Virginia Tech **November 2025 - Present**
Lead Trajectory and Aerodynamics Engineer
- Led an interdisciplinary engineering subteam in aerodynamic analysis, CFD modeling (ANSYS Fluent), and trajectory simulations to evaluate flight performance, aerodynamic loads, and stability across supersonic flight regimes.
- Improved vehicle static margin by 10% through fin and nosecone optimization; redesigned fin geometry to achieve a simulated apogee 15% above the Kármán line target using RASAero II.

NASA L'SPACE Mission Concept Academy **May 2026 - September 2026**
Payload Mechanical Engineer
- Designed lunar rover mechanical components and subsystem layouts in Siemens NX, conducting suspension, wheel, and structural trade studies to evaluate mass, manufacturability, thermal performance, and terrain capability.
- Coordinated interdisciplinary subsystem development, translating mission objectives into mechanical requirements, interface definitions, verification criteria, and design decisions through preliminary design review.

Projects

Aurora (2D Subsonic Airfoil Analysis Tool) **June 2026 - Present**
- Developed a Python-based aerodynamic solver integrating a vortex panel method, boundary layer modeling, and iterative viscous–inviscid coupling to predict airfoil lift, drag, pitching moment, and pressure distributions.
- Implemented automated aerodynamic polar generation, multi-airfoil comparison, and data visualization; validated aerodynamic predictions against AirfoilTools reference data.